Furuno DRS4D-NXT Radar

Wire Protocol & System Technical Manual

J.C. Vaught

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A complete technical reference for the Furuno DRS4D-NXT marine radar system, covering the wire protocol, echo data encoding, system architecture, web interface, and development guide.

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Part I

Overview & Usage

1 Introduction

This document is the complete technical reference for the Furuno DRS4D-NXT marine radar system as interfaced through the `furuno-server` Rust application. It covers radar operation via a browser-based web interface, the internal software architecture, echo data encoding and decoding, and the full wire protocol. The intended audiences are new users learning to operate the system, contributors modifying the codebase, and developers building their own applications on top of the radar hardware.



Figure 1: System overview. The radar scanner communicates with the Rust server over a dedicated Ethernet link. The server exposes a web interface accessible from any browser on the network.

1.1 Hardware Overview

The Furuno DRS4D-NXT is a solid-state Doppler marine radar operating in the X-band at 9.41 GHz. The scanner unit contains the antenna, transmitter, and receiver in a single sealed radome enclosure. It connects to a host processor via a standard Ethernet cable. The scanner is assigned a fixed IP address of `172.31.3.226` on a `172.31.0.0/16` subnet.

The radar transmits at up to 25 watts using solid-state amplification rather than a traditional magnetron, which eliminates warm-up time and provides consistent performance across the operating temperature range. The antenna rotates at either 24 or 48 RPM depending on the selected speed mode. Echo data is delivered as radial spokes, with 8192 unique azimuth positions per full rotation and up to 868 range bins per spoke. The system supports six pulse width settings (S1, S2, M1, M2, M3, L), 18 range scales from 1/16 NM to 48 NM, and built-in Doppler target discrimination.

1.2 Software Components

The system consists of three software components.

furuno-server is a Rust binary that implements the complete Furuno wire protocol from scratch. It handles network discovery, authentication, control commands, echo data reception and decoding across four encoding types, PPI (Plan Position Indicator) rendering, and serves a web interface for browser-based operation. The application is built with Axum 0.7 for the HTTP server and Tokio

for the async runtime. All radar communication—TCP control, UDP multicast echo reception, and subnet management—is handled natively without any Furuno proprietary libraries.

The *web interface* is a set of static HTML, CSS, and JavaScript files served by the Rust application. It provides four pages: a controls dashboard for quick radar operation, a full-screen PPI display with history compositing, a detailed settings panel, and a system status view. The interface communicates with the server through a REST API and polls for updated radar images at 500 ms intervals.

The *documentation* consists of this technical manual and inline source code comments. The manual is maintained as a single self-contained LaTeX file with all diagrams rendered via TikZ.

2 Quick Start & Installation

2.1 Prerequisites

The Rust toolchain is required to build the application. Install it via `rustup` if it is not already present.

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source ~/.cargo/env
```

The application builds and runs on Linux (x86_64 and aarch64), macOS, and Windows. No external C libraries or Furuno SDKs are required.

2.2 Building

From the repository root, build the release binary.

```
cd furuno-server
cargo build --release
```

The compiled binary is located at `furuno-server/target/release/furuno-server`. A release build is recommended for deployment as it enables compiler optimizations that significantly improve echo decoding and PPI rendering throughput.

2.3 First Run

Start the web server with the `web` mode.

```
cargo run --release -- web
```

The web interface is then available at `http://localhost:8080` in any browser. On startup, the application attempts to discover the radar automatically (see below). If the radar is not reachable, the web server starts anyway and serves a degraded interface. The `/reconnect` endpoint can be used to retry the connection later.

2.4 Auto-Discovery

The application resolves the radar and host IP addresses through a three-stage fallback process.

First, it checks for explicit command-line flags. If `--radar-ip` is provided, that address is used directly. The `--host-ip` flag is optional; if omitted, the application determines a suitable host address on the radar's subnet automatically.

```
cargo run --release -- --radar-ip 172.31.3.226 web
```

Second, if no flags are provided, the application broadcasts UDP discovery beacons on port 10010 across all detected network interfaces and listens for 3 seconds. If the radar responds, both the radar and host addresses are extracted from the response.

Third, if discovery fails, the application falls back to the default addresses: `172.31.3.226` for the radar and `172.31.1.100` for the host. If the host does not already have an interface on the `172.31.0.0/16` subnet, a secondary IP alias is added to the physical Ethernet interface automatically. On Linux this uses `ip addr add`; on macOS, `ifconfig alias`.

3 CLI Modes

3.1 General Syntax

The application accepts optional IP flags followed by a mode name and mode-specific arguments.

```
furuno-server [--radar-ip IP] [--host-ip IP] <mode> [args...]
```

The two user-facing modes are `web` and `discover`. Additional developer-oriented modes (`handshake`, `tx-on`, `tx-off`, `echo`, `full`, `render-live`, `dump-raw`, `test-commands`, `validate`) are documented in §13. If no mode is given, the application defaults to `handshake`.

3.2 Web Server Mode

The `web` mode starts the HTTP server and is the primary way to operate the radar.

```
furuno-server web [port] [webdir]
```

The optional `port` argument sets the listening port (default 8080). The optional `webdir` argument specifies the path to the static web files (default `../web` relative to the binary). On startup, the server attempts to authenticate with the radar and initialize the control channel. If the radar is unreachable, the server starts anyway with degraded functionality—the PPI display shows a blank image and setting changes are queued until a connection is established via the `/reconnect` endpoint.

3.3 Discover Mode

The `discover` mode broadcasts UDP beacons on port 10010 across all network interfaces and prints any radar responses it receives.

```
furuno-server discover
```

This is useful for determining the radar's IP address on an unfamiliar network or verifying that the radar is powered on and reachable. The mode runs until a response is received or the timeout expires.

4 Web Interface

4.1 Starting the Server

The web server starts when the application is launched in `web` mode (see §3.2). By default it listens on port 8080 and serves static files from the `../web` directory relative to the binary. Both values can be overridden with positional arguments. The server starts even if the radar is not reachable, allowing the interface to load and wait for a connection. The `/reconnect` API endpoint triggers a fresh authentication and initialization attempt without restarting the server.

4.2 Page Tour

The web interface consists of four HTML pages, all sharing a common header with navigation links, a consistent CSS stylesheet, and a single shared JavaScript file (`script.js`).

4.2.1 Controls (index.html)

The controls page is the default landing page and provides the most commonly used radar operations. It is divided into two panels. The left panel displays a radar preview image (refreshed every 2 seconds) with a link to the full-screen radar view. The right panel contains quick controls for TX power (on/off), CSV recording (start/stop), range selection (17 scales from 0.0625 to 36 NM), pulse width (S1 through L), manual gain (0–100), and RezBoost level (off/low/medium/high). Each control has an apply button and a refresh button that queries the radar's current state.

Below the quick controls is a pattern loader panel. Users can load a pattern file (`.txt` or `.csv`), set a repeat count (-1 for infinite looping), assign an optional label, and add the pattern to a queue. The queue is displayed as an ordered list, and patterns execute sequentially when the RUN button is pressed. A STOP button halts execution mid-pattern, and RESET clears the queue.

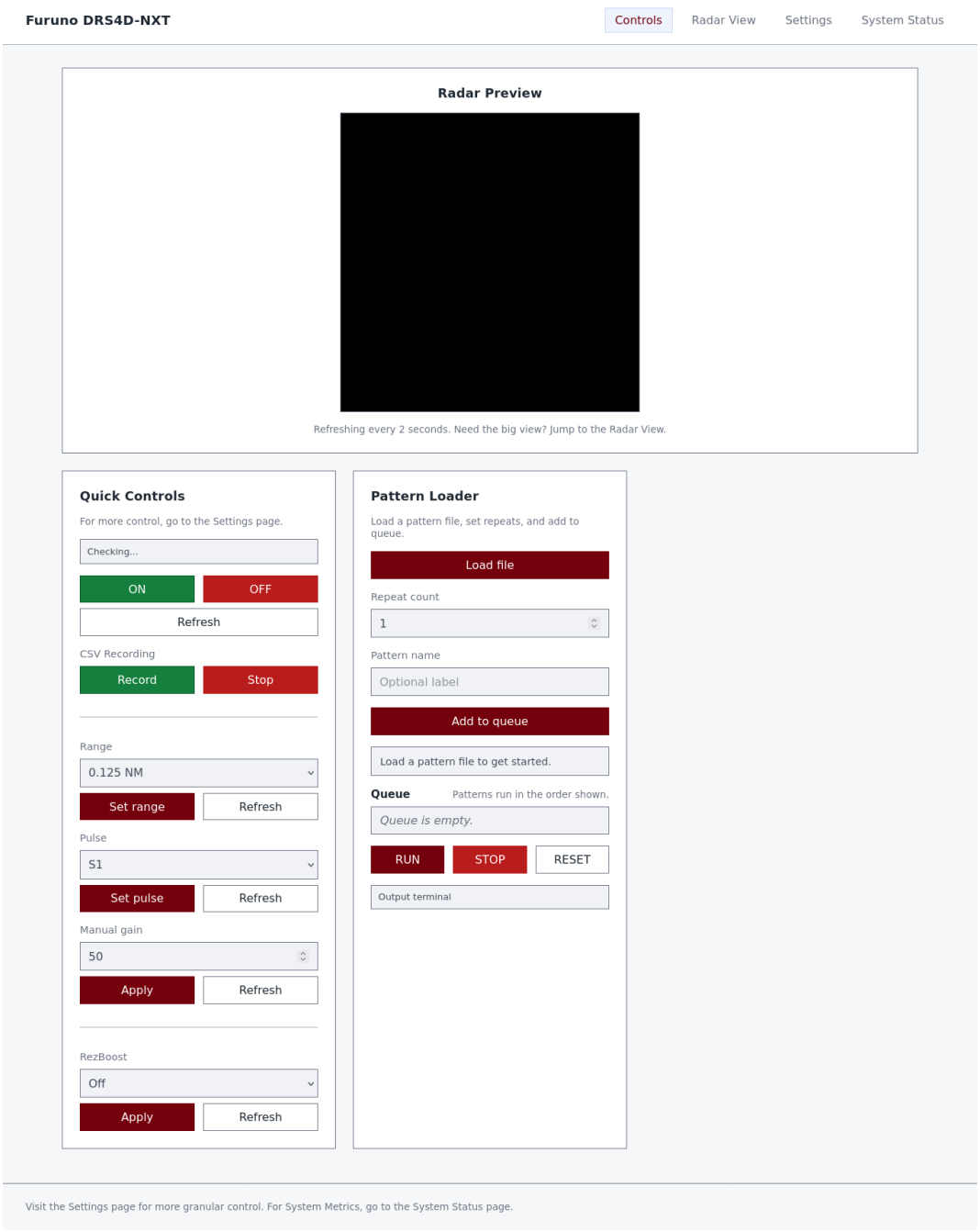


Figure 2: Controls page (`index.html`). The radar preview is shown at top, with quick controls and the pattern loader below.

4.2.2 Radar View (radar.html)

The radar view page provides a full-screen PPI display. The primary display element is an HTML5 canvas (1024×1024 pixels) that composites multiple frames of radar data to show target history. Below the canvas are controls for track length (1–50 frames), current frame color, and track history color, with a reset button to restore defaults. An `<img>` element behind the canvas serves as a

fallback if canvas rendering is unavailable. The canvas is refreshed every 500 ms by fetching the latest PPI image from the `/newest_png` endpoint.

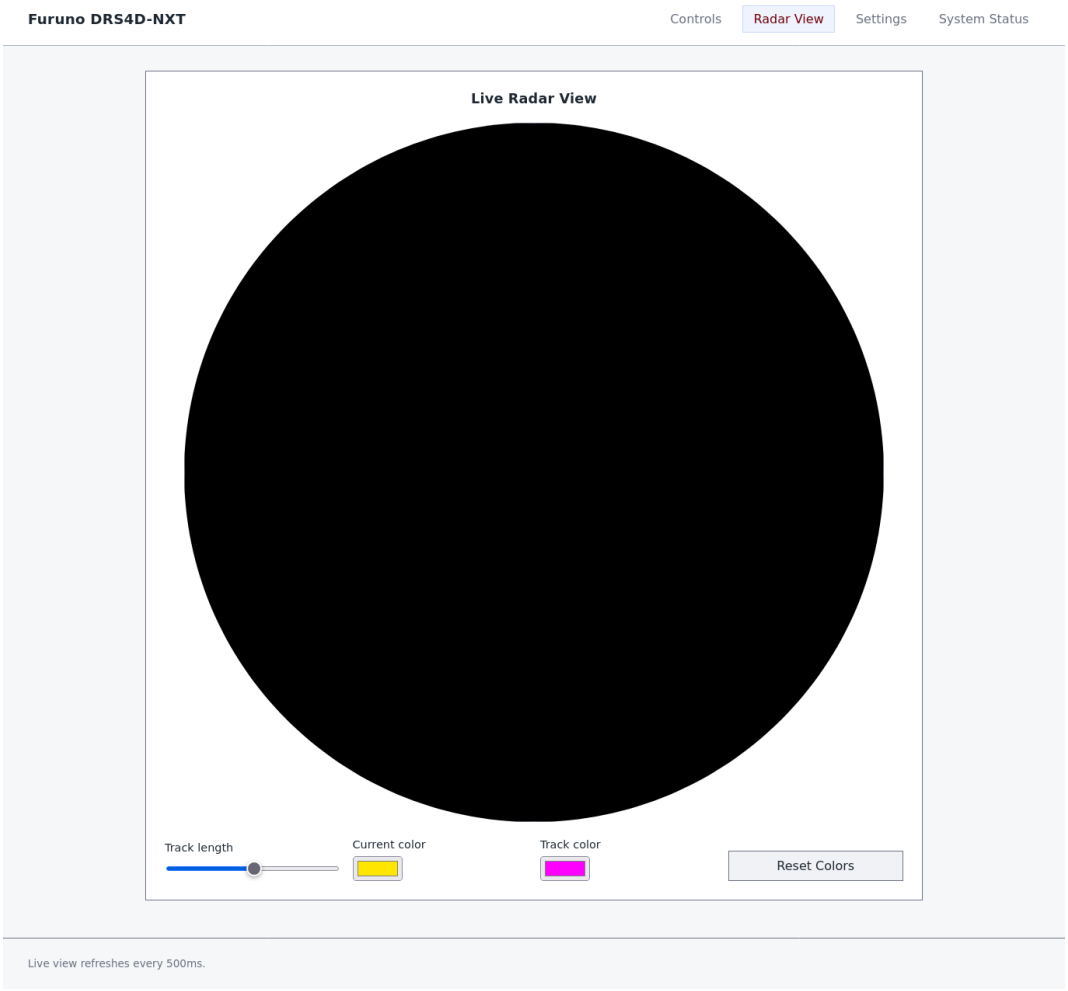


Figure 3: Radar view page (`radar.html`). The PPI display with history compositing and track color controls.

4.2.3 Settings (settings.html)

The settings page provides granular control over all radar parameters. It is organized into panels grouped by function.

The *connection panel* shows the current connection status and provides reconnect and refresh buttons. A small radar preview image is displayed alongside.

The *power control panel* provides TX on/off buttons and a refresh button to query the current transmitter state.

The *range and pulse panel* allows independent or combined adjustment of range and pulse width, with the same options available on the controls page.

The *gain panel* supports both manual and automatic gain modes. In manual mode, the user sets

a value from 0 to 100. In auto mode, a bias offset (0–100) adjusts how aggressively the automatic gain control responds.

The *STC panel* exposes three sensitivity time control zones—near, mid, and far—each with an independent numeric input.

The *signal processing panel* groups main bang suppression (0–255), interference rejection (on/off), bird mode (off/low/medium/high), and Doppler target analysis (off, internal heading, external heading).

The *clutter panel* provides sea clutter (A/C SEA) and rain clutter (A/C RAIN) controls, each with manual and automatic modes. Manual mode sets a fixed rejection level; auto mode adjusts a bias for the radar's internal adaptive filter.

The *antenna speed panel* controls rotation speed (auto/low/high) and a high-RPM override toggle.

The *RezBoost and display panel* sets the resolution enhancement level and display parameters including sweep line count and interpolation mode.

The *pattern loader* on this page has the same functionality as the one on the controls page, with the addition of a manual text input option for entering pattern steps directly.

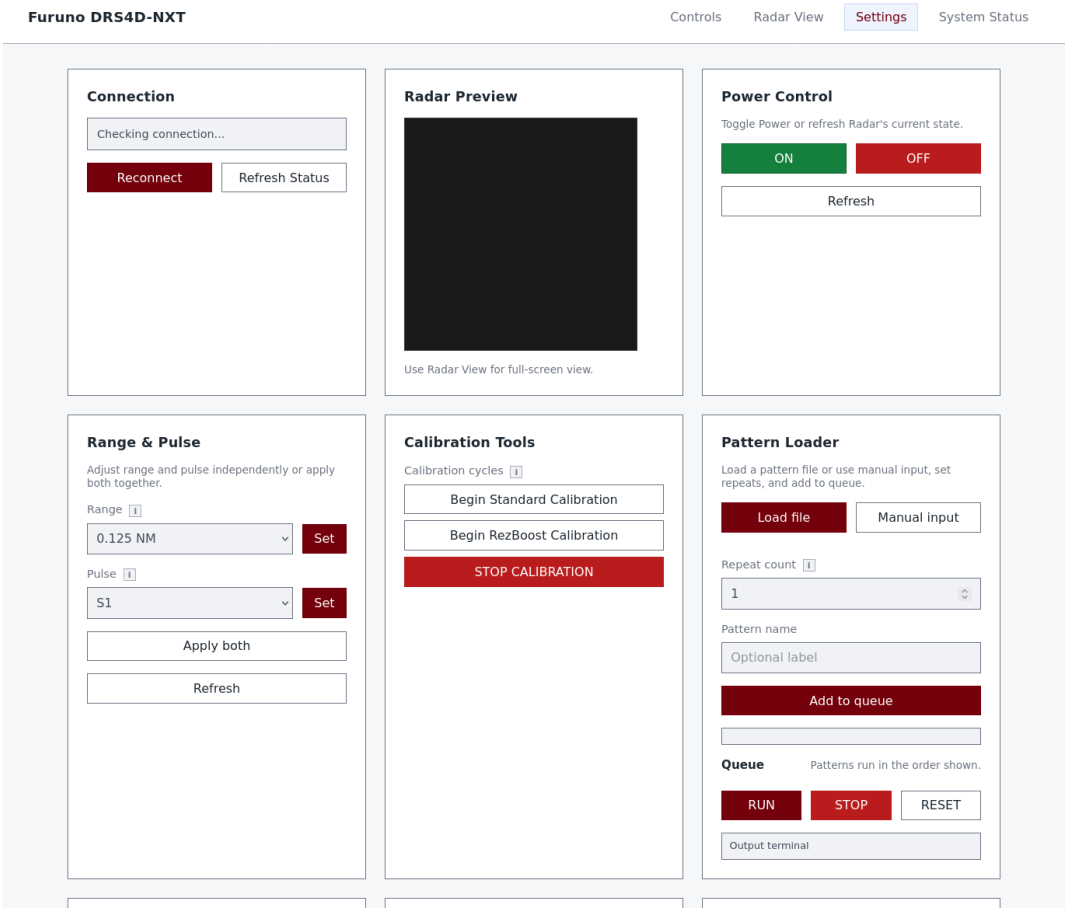


Figure 4: Settings page (`settings.html`), upper half. Connection, radar preview, power control, range/pulse, calibration tools, and pattern loader panels.

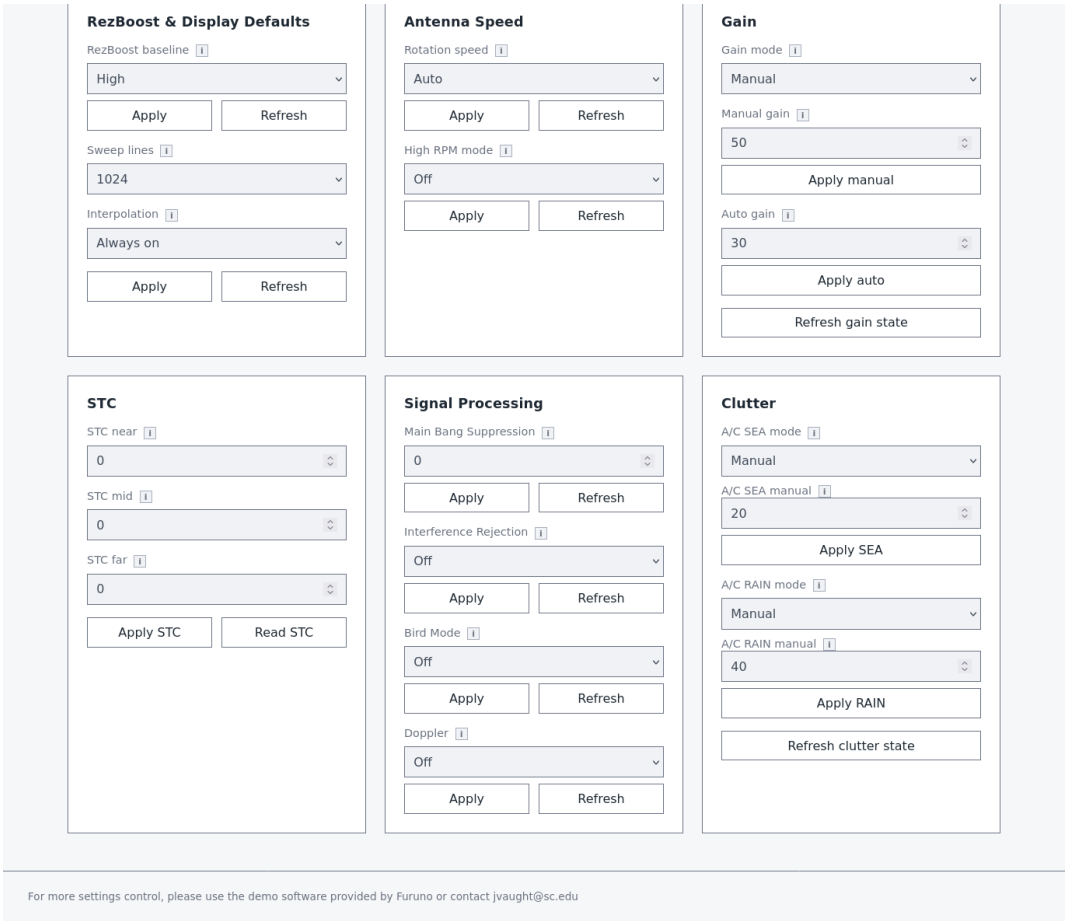


Figure 5: Settings page (`settings.html`), lower half. RezBoost, antenna speed, gain, STC, signal processing, and clutter panels.

4.2.4 System Status (status.html)

The system status page displays live telemetry from both the radar and the host system, updated every 500 ms.

The *radar connection* panel shows connection status, radar IP, host IP, TX state, and firmware version. The *session* panel shows server uptime, total rotations received, recording state, the current recording directory, and pattern queue status. The *active settings* panel displays a read-only summary of all current radar parameters: range, pulse, gain, RezBoost, antenna speed, Doppler, interference rejection, bird mode, main bang suppression, STC values, and clutter settings.

The *host resources* panels display CPU load (overall and per-core), memory usage, storage usage, network throughput, and CPU temperature. Each metric includes a progress bar and numeric label.

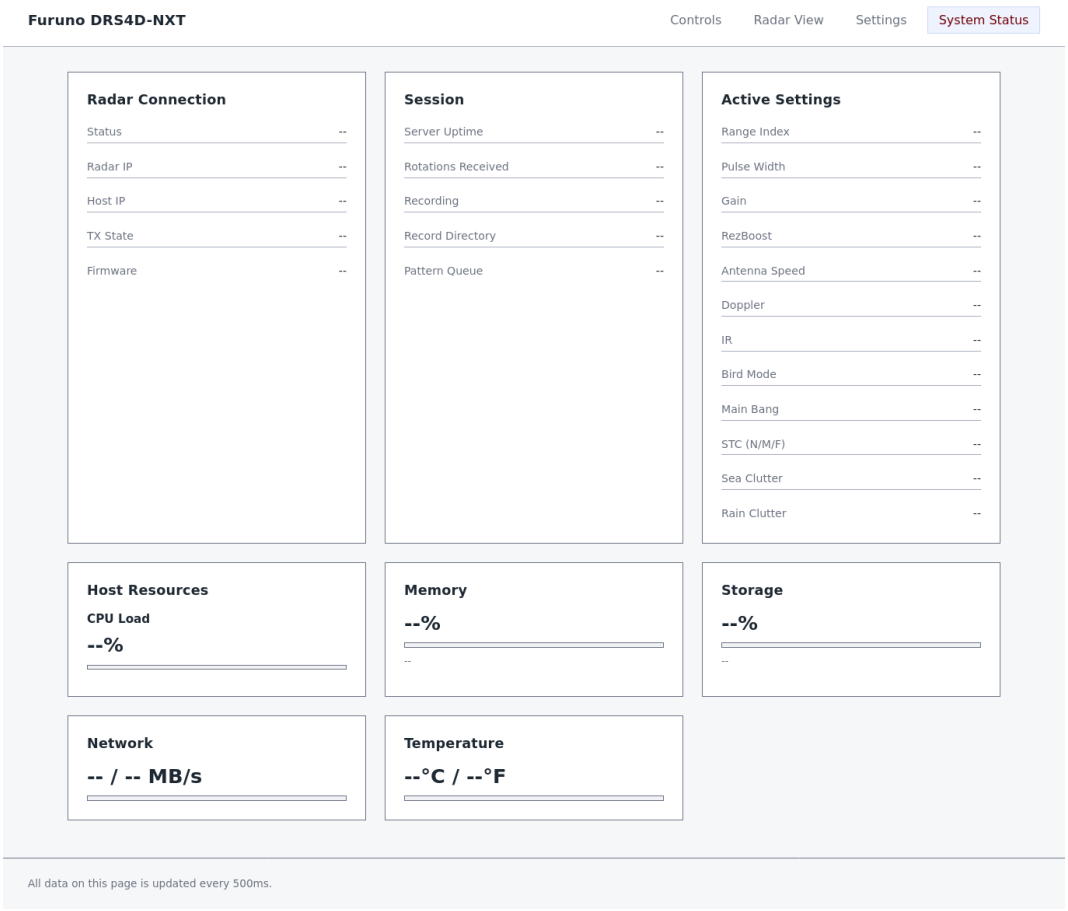


Figure 6: System status page (`status.html`). Live telemetry from the radar and host system.

4.3 PPI Display

The PPI (Plan Position Indicator) display renders radar echo data as a top-down circular image with the radar at the center. The server renders each complete antenna rotation into a 640×640 pixel PNG image using the warm-tinted color mapping described in §8.2. The browser fetches the latest image via the `/newest_png` endpoint.

On the radar view page, the JavaScript composites multiple frames onto an HTML5 canvas to show target motion history. The *track length* slider controls how many previous frames are retained (1–50). The *current color* picker tints the most recent frame, and the *track color* picker tints older frames. Older frames are drawn with decreasing opacity to produce a fading trail effect. This compositing runs entirely in the browser; the server delivers only single-frame PNG images.

5 Recording & Data Formats

5.1 Starting and Stopping Recording

Recording is toggled through the web interface or by calling the `/toggle_recording` API endpoint with a `state` parameter of `start` or `stop`. When recording starts, the server creates a new timestamped directory under `recordings/` with `csv/` and `png/` subdirectories. If a directory with the same timestamp already exists, a numeric suffix is appended (e.g., `_01`, `_02`). Each complete antenna rotation produces one CSV file and one PNG file, written automatically by the echo listener thread. When recording is stopped, a `settings.json` snapshot is written to the recording directory.

5.2 Output Directory Structure

Each recording session produces the following directory structure.

```
recordings/<timestamp>/
  csv/rotation_<ts>.csv
  png/rotation_<ts>.png
  settings.json
```

The top-level directory is named with a compact timestamp (e.g., `20260307_143022`). For pattern-based recordings, the sanitized pattern label is appended (e.g., `20260307_143022_sweep1`). The `csv/` and `png/` subdirectories are created on demand when the first rotation is written.

5.3 CSV Format

Each CSV file represents one complete antenna rotation. The file begins with a metadata header of `#`-prefixed comment lines that capture all radar settings at the time of recording.

```
# range_index,4
# gain_auto,false
# gain_value,50
# pulse_width,0
# sea_clutter,0
# rain_clutter,0
# rezboost,0
# interference_rejection,false
# bird_mode,0
# doppler,0
# antenna_speed,0
# main_bang,0
# near_stc,0
# middle_stc,0
# far_stc,0
Status  Scale   Range   Gain    Angle   EchoValues
0   4   4   19  0   12,14,8,...
0   4   4   19  1   10,11,9,...
```

After the header, the column line and data rows are tab-separated. Each row represents one spoke and contains a status field (always 0), the range scale index, the range index, the gain percentage (mapped from the 0–255 wire range to 0–100), the azimuth angle in ticks (0–8191), and the echo bin values as a comma-separated list of integers.

5.4 Settings Snapshot

When a recording session ends, the server writes a `settings.json` file to the recording directory. The snapshot format depends on how the recording was initiated.

For manual recordings, the snapshot contains the mode (`"manual"`), the recording directory path, the full `CurrentSettings` object, and a `history` array of `SettingChange` entries. Each history entry records a timestamp, the setting name, and the new value, capturing every parameter change made during the recording session.

For pattern-based recordings, the snapshot additionally includes the `PatternInfo` object describing the pattern that was executing, including all pattern steps with their parameter values and repeat counts.

6 Pattern System

6.1 What Patterns Do

Patterns automate sequences of radar parameter changes. A pattern consists of one or more *steps*, each specifying a set of radar settings and a *repeat count* that determines how many full antenna rotations that step is held before advancing to the next. When recording is active during pattern execution, each rotation produces a CSV and PNG file, allowing systematic data collection across parameter combinations.

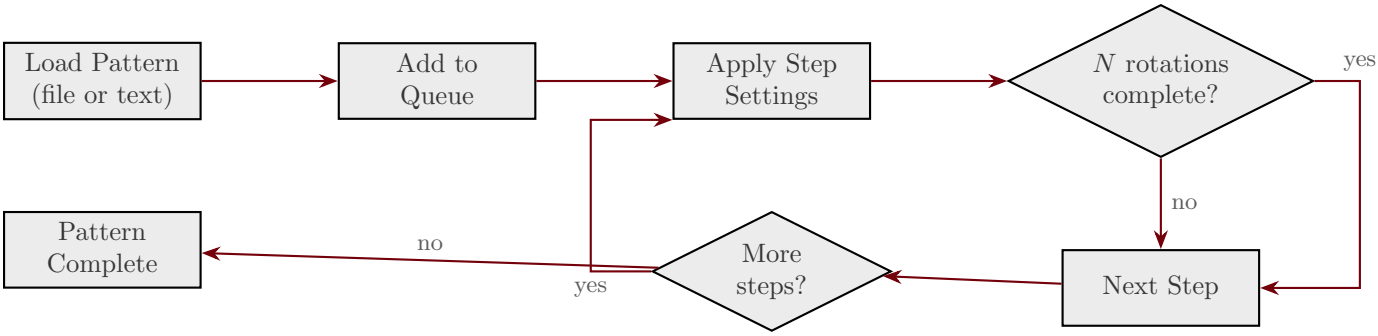


Figure 7: Pattern execution flow. Each step is applied to the radar and held for the specified number of antenna rotations before advancing.

6.2 Simple Pattern Format (7-Field)

The simple format uses seven comma-separated fields per step. All fields are required and all values are set explicitly.

```
range, gain, pulse, rezboost, speed, interp, repeat
```

For example, the step `4,50,0,3,2,1,5` sets range index 4 (1 NM), gain 50, pulse S1, RezBoost high, auto antenna speed, interpolation on, and holds for 5 rotations. Multiple steps are separated by semicolons.

```
4,50,0,3,2,1,5; 8,50,2,3,2,1,5; 11,50,4,3,2,1,5
```

6.3 Advanced Pattern Format (13-Field)

The advanced format extends the simple format with six additional fields for fine-grained signal processing control.

```
range, gain, pulse, rezboost, speed, interp,
    sea_clutter, rain_clutter, main_bang,
    interference_rejection, bird_mode, doppler, repeat
```

Any of the 12 parameter fields (all except `repeat`) can be set to `NaN` to leave that setting unchanged from its current value. This is particularly useful when only a subset of parameters needs to vary across steps. For example, the step `4,NaN,0,NaN,NaN,NaN,NaN,NaN,NaN,NaN,NaN,NaN,NaN,3` changes only the range to index 4 and the pulse to S1, leaving all other settings at their current values and holding for 3 rotations.

6.4 Pattern Queue

Multiple patterns can be loaded into a queue and executed sequentially. Each queue entry has its own pattern data, repeat count, and optional label. The repeat count controls how many times the entire pattern is cycled (-1 for infinite looping). Patterns execute in the order they were added.

During queue execution, each pattern item creates its own recording session with a separate timestamped directory (tagged with the sanitized pattern label if one was provided). The queue can be stopped at any time using the STOP button, which halts execution between steps. A RESET button clears the entire queue.

6.5 Loading Patterns

Patterns can be loaded in two ways. The first is file upload: clicking the "Load file" button opens a file picker that accepts `.txt` or `.csv` files. The file contents are parsed as semicolon-separated steps. The second is manual input: on the settings page, clicking "Manual input" reveals a text area where steps can be typed or pasted directly.

After loading, the user sets a repeat count and an optional label, then clicks "Add to queue" to append the pattern to the queue list. Multiple patterns can be queued before pressing RUN.

6.6 NaN Override Behavior

When a pattern step uses `NaN` for a field, that setting remains *unlocked* during execution. This means the user can manually change that parameter through the web UI while the pattern runs, and the change takes effect immediately. Conversely, parameters with explicit values in the current step are *locked*—manual changes to locked parameters are rejected with an error message indicating that the setting is controlled by the active pattern.

This mechanism allows hybrid operation where some parameters are swept by the pattern while others are adjusted interactively. For example, a pattern that sweeps range and pulse but sets all clutter fields to `NaN` lets the operator tune clutter rejection in real time while the automated sweep runs.

Part II

Software Internals

7 System Architecture Overview

This section provides a high-level map of the codebase. Each subsystem is introduced briefly here and covered in depth in subsequent sections.

7.1 Module Summary

The Rust server is organized into the following source modules, all compiled as a single binary via `main.rs`.

File	Role	Details
`main.rs`	Entry point	CLI argument parsing, IP resolution, mode dispatch
`protocol.rs`	Wire protocol	Packet encoding/decoding, command builders, echo decompression
`connection.rs`	State machine	TCP auth, control channel, keepalives, response parsing (§14)
`web.rs`	Web server	Axum HTTP handlers, echo listener thread, recording, pattern engine (§9–§12)
`echo.rs`	Echo helpers	Rotation collection, CSV export, echo header parsing
`render.rs`	PPI renderer	Polar-to-Cartesian projection, color mapping, PNG encoding (§8)
`modes.rs`	CLI modes	Standalone operations: handshake, TX control, dump, render-live (§13)
`testing.rs`	Test harness	A/B command verification, 6-phase validation suite (§15)
`tests.rs`	Unit tests	`#[cfg(test)]` integration tests requiring live radar

Table 1: Source module summary.

7.2 System-Level Block Diagram

The following diagram shows the physical and network boundaries of the system. The radar hardware communicates with the Rust server over a dedicated Ethernet link, and the server exposes an HTTP interface to the browser.

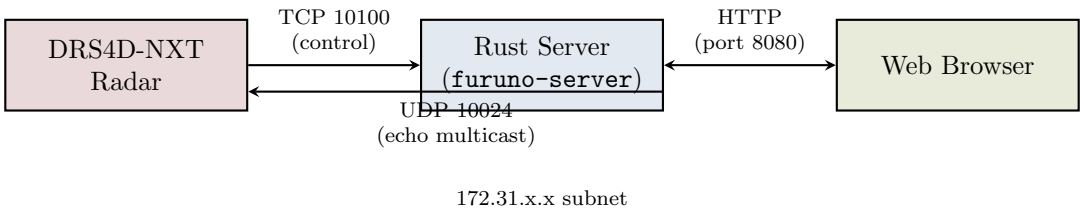


Figure 8: System-level block diagram showing radar hardware, Rust server, and browser.

7.3 Module-Level Diagram

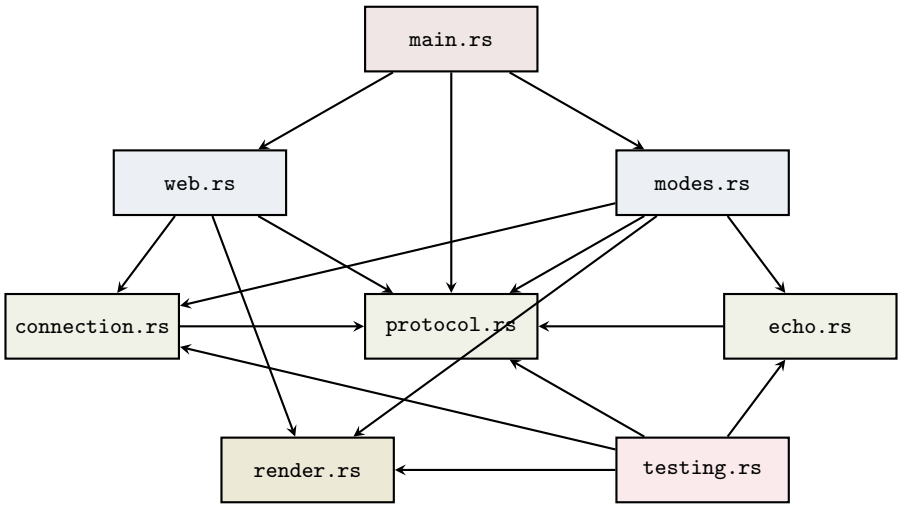


Figure 9: Module dependency diagram. Arrows point from dependent module to dependency.

7.4 Data Flow Diagram

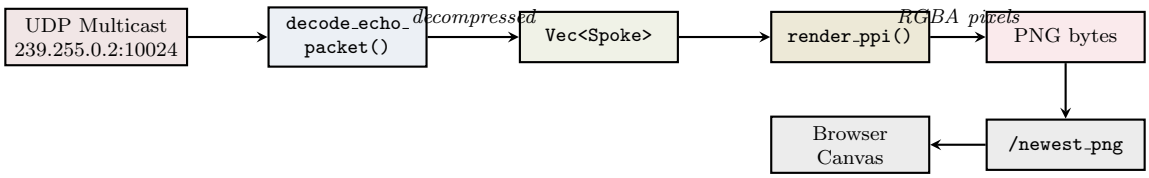


Figure 10: Echo data flow from UDP multicast to browser canvas.

7.5 Concurrency Model

The server uses a hybrid concurrency model combining Tokio async tasks with a dedicated OS thread.

The **Tokio async runtime** handles all HTTP request processing. Each Axum handler runs as an async task on the Tokio thread pool. Blocking radar operations (sending commands, waiting for acknowledgments) are wrapped in `tokio::task::spawn_blocking()` to avoid blocking the async executor.

A single **dedicated** `std::thread` runs the echo listener loop. This thread receives UDP multicast packets, decodes spokes, renders the PPI image, and optionally writes CSV/PNG files during recording. It runs independently of the HTTP handlers and communicates with them through shared state.

Synchronization primitives in the shared `AppState` struct coordinate access between the async handlers and the echo thread. `Mutex<RadarConnection>` provides exclusive access to the TCP control channel, ensuring only one thread sends commands at a time. `RwLock<Vec<u8>>` protects the latest PNG image, allowing the echo thread to write and multiple HTTP handlers to read

concurrently. Several `AtomicBool` flags (`tx_active`, `recording`, `pattern_stop`) provide lock-free coordination for frequently checked boolean state.

8 PPI Rendering

The PPI (Plan Position Indicator) renderer in `render.rs` converts a vector of decoded `Spoke` structs into a 640×640 pixel RGBA image. This is the most commonly modified subsystem when changing visual appearance, image resolution, or overlay elements.

8.1 Coordinate Transformation

Each spoke has an `angle_ticks` field representing a compass bearing (0 = north, increasing clockwise) in the range 0–8191 (8192 ticks per full rotation). The renderer converts this to a standard math angle (0 = east, increasing counter-clockwise) using the formula:

$$\theta_{\text{rad}} = \frac{\texttt{angle_ticks}}{8192} \times 2\pi \tag{1}$$

$$\theta_{\text{math}} = \frac{\pi}{2} - \theta_{\text{rad}} \tag{2}$$

For each spoke, the renderer walks outward from the image center along the radial line defined by θ_{math}, one pixel step at a time. At each step at radius r, the corresponding range bin index is calculated by linear interpolation:

$$\texttt{bin} = \left\lfloor \frac{r}{r_{\max}} \times N_{\text{bins}} \right\rfloor \tag{3}$$

where $r_{\max} = 318$ pixels (center minus 2-pixel margin) and $N_{\text{bins}} = 868$ (the standard number of range bins per spoke). The pixel coordinates are:

$$p_x = \texttt{center} + r \cdot \cos(\theta_{\text{math}}) \tag{4}$$

$$p_y = \texttt{center} - r \cdot \sin(\theta_{\text{math}}) \tag{5}$$

8.2 Color Mapping

Echo amplitudes arrive as 6-bit values (0–63) after the `wire >> 2` display conversion. The renderer scales these to 8-bit intensity and applies a warm-tinted color mapping:

$$\texttt{intensity} = \min\left(\frac{\texttt{amplitude} \times 255}{63},\ 255\right) \tag{6}$$

$$R = \texttt{intensity} \tag{7}$$

$$G = \left\lfloor \texttt{intensity} \times \frac{180}{255} \right\rfloor \approx 0.706 \times \texttt{intensity} \tag{8}$$

$$B = \left\lfloor \texttt{intensity} \times \frac{20}{255} \right\rfloor \approx 0.078 \times \texttt{intensity} \tag{9}$$

This produces a warm amber/orange color for strong returns, fading to dark red/brown for weak returns. Zero-amplitude bins are left as the black background (RGBA 0,0,0,255).

8.3 Overlay Elements

After all spokes are drawn, the renderer adds two overlay elements in dark gray (RGB 54, 54, 54).

Crosshairs are drawn as a horizontal and vertical line passing through the image center, spanning the full image width and height.

Range rings are drawn as four equally-spaced circles at radii $r_{\max}/4$, $r_{\max}/2$, $3r_{\max}/4$, and $r_{\max}$. Each ring is rasterized by stepping around its circumference with angular resolution proportional to its perimeter.

8.4 PNG Encoding

Two functions handle PNG output, both using the `png` crate.

`encode_png_bytes(pixels, width, height)` encodes the RGBA pixel buffer into an in-memory `Vec<u8>` containing the complete PNG file. This is used by the echo listener thread to update the `latest_png` shared state without touching the filesystem.

`write_png(path, pixels, width, height)` writes the PNG directly to a file via a buffered writer. This is used during recording (per-rotation PNG saves) and by CLI modes that produce output files.

Both functions encode as RGBA color type with 8-bit depth.

8.5 How to Modify

Image size. Change the `PPI_SIZE` constant (currently 640). All geometry is computed relative to this value, so no other changes are needed. Larger sizes produce smoother radial lines at the cost of increased memory and PNG encoding time.

Color palette. Modify the three lines in `render_ppi()` that assign to `pixels[idx]`, `pixels[idx+1]`, and `pixels[idx+2]`. For a green phosphor look, use $R = 0$, $G =$ `intensity`, $B = 0$. For a blue-white scheme, use $R =$ `intensity`$/2$, $G =$ `intensity`$/2$, $B =$ `intensity`.

New overlays. Add drawing code after the range ring loop in `render_ppi()`. The pixel buffer is a flat `Vec<u8>` in row-major RGBA order. The index for pixel (x, y) is $(y \times \texttt{size} + x) \times 4$, with offsets +0 for R, +1 for G, +2 for B, +3 for A.

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